

Data Structure and Algorithms

CSIT — 3rd Semester — 2081 — Answer Sheet
Subject Code: CSC211

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Compute the GCD of 30 and 12 using recursion. Convert the infix expression $A + B * (C - D) / E$ to postfix using stack.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Explain circular linked list in brief. Explain process of inserting and removing nodes in a singly linked list.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain merge sort along with its time complexity. Trace the execution of Quick sort algorithm with the array of numbers 40, 22, 70, 11, 43, 18, 20, 21 and 8.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Why do we need dynamic memory allocation? Explain abstract data type with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is drawback of linear queue? Explain circular queue.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. List any one limitation of recursion. Describe stack as linked list.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define max and min heap. How can we use heap property in heap sort? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write the algorithm of binary search with its time complexity.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Suppose, the set of keys is {5, 10, 12, 8, 47, 56, 7, 48}, $m = 10$, and $h(x) = x \bmod 10$. Show the effect of successively inserting these keys using linear probing.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Create a BST from data 20, 25, 18, 16, 19, 30, 17 and traverse it in inorder, preorder and post order.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. How do you represent graphs? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on: a) Priority queue b) Minimum spanning tree

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.